

# **Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based**

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## METHODOLOGY

### 1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

| Parameter          | Sentinel-1 C-SAR (IW Mode)      |
|--------------------|---------------------------------|
| Frequency Band     | C-band (~5.405 GHz)             |
| Revisit Time       | 12 days (single satellite)      |
| Spatial Resolution | 5 m (range) × 20 m (azimuth)    |
| Coverage Mode      | Interferometric Wide Swath (IW) |

### 2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

|                                                                     |
|---------------------------------------------------------------------|
| S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D |
| S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431 |
| S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6 |
| S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312 |
| S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14 |
| S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F |
| S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445 |
| S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52 |
| S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD |
| S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8 |
| S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13 |
| S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2 |
| S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5 |
| S1A_IW_SLC__1SDV_20250616T130237_20250616T130306_059672_0768D1_F57D |
| S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD |

### 3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



## VISUALIZATIONS

### 1. Defining specific bridge location & parameters for satellite SAR based inspection



#### Project & Bridge Details

State – Rajasthan

Bridge No. – 899/1

Bridge Chainage – 1032+420

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – MNB

#### Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

Flight Direction - DESCENDING

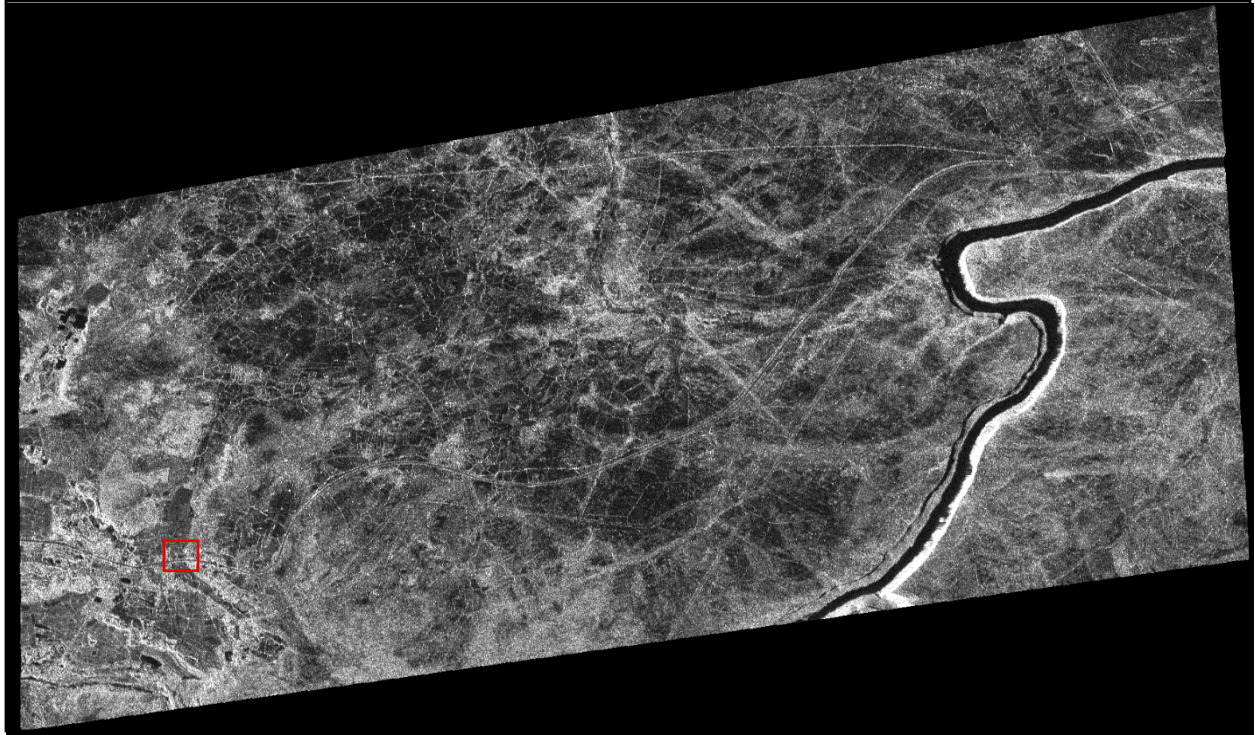
Polarization – VV+VH

Absolute Orbit – 54706

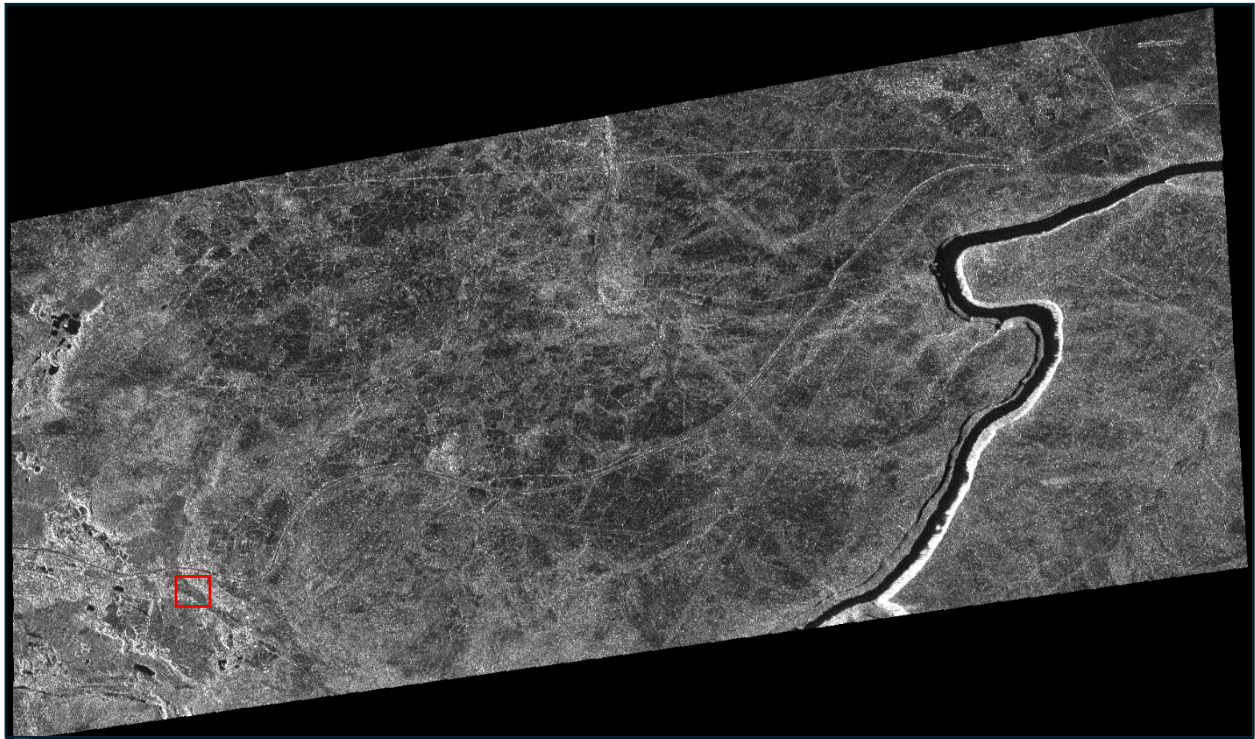


## 2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization





### 3. Results of SAR based bridge inspection, including visualizations

#### Bridge Components

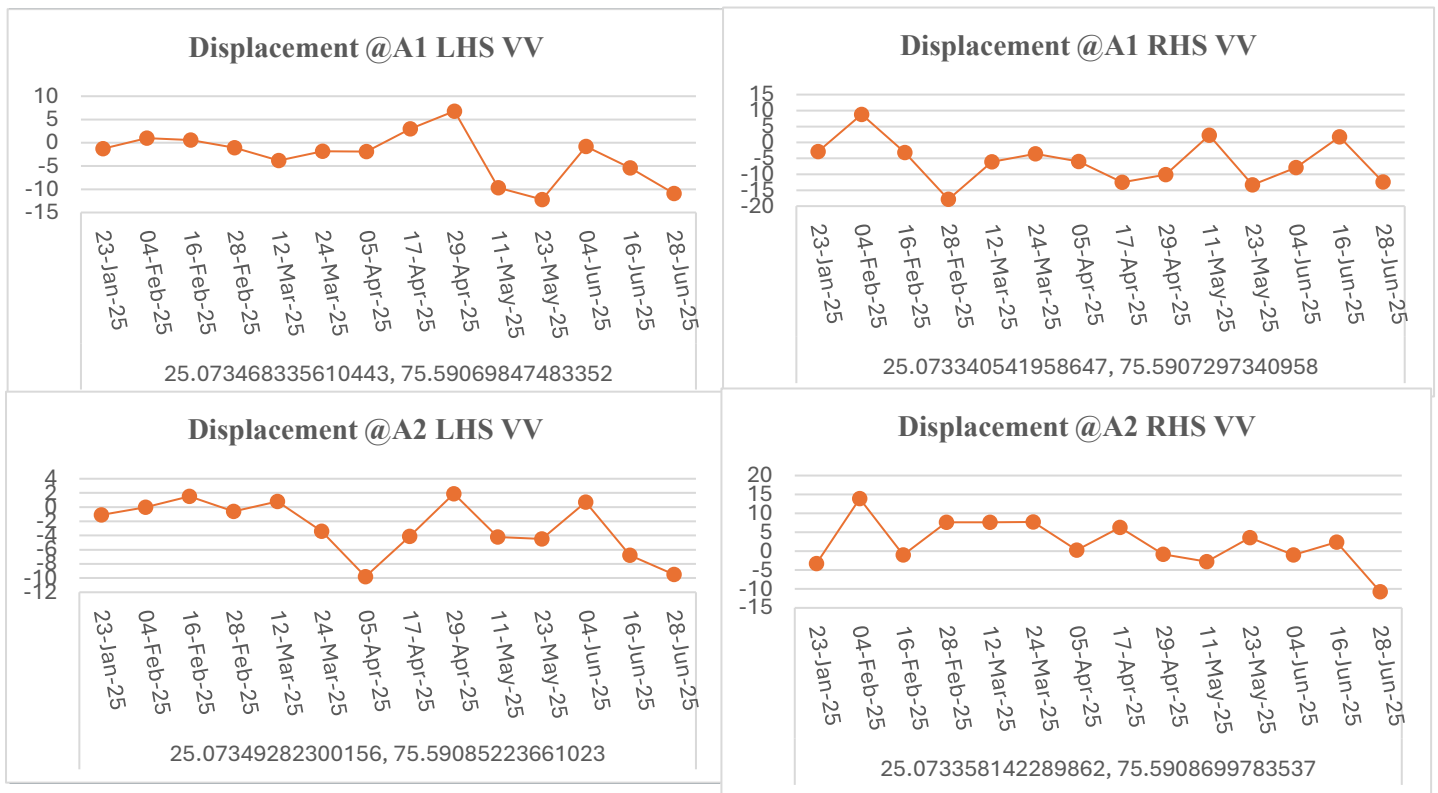
MNB – CH-1032+420

Length – 25.18M

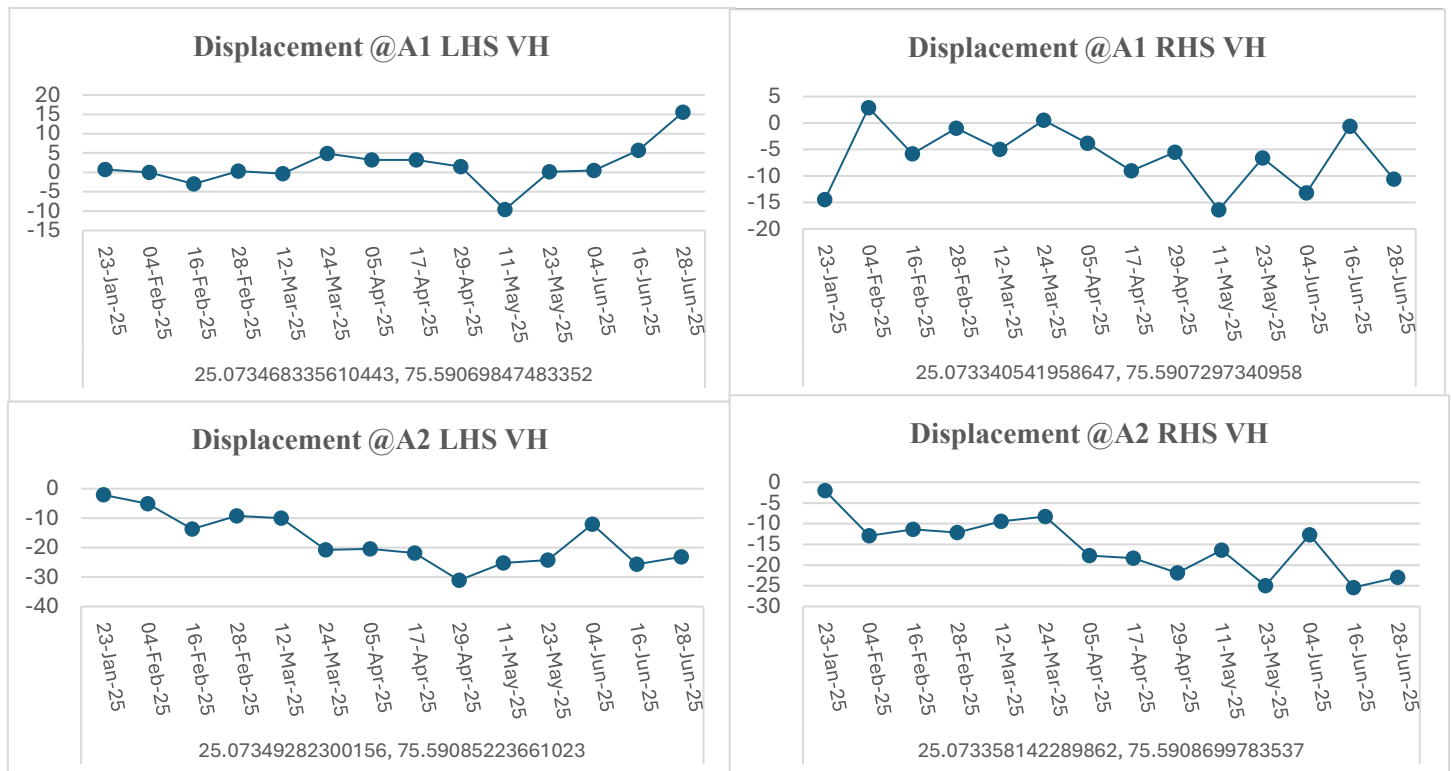


|                                      |  |                    |                   |
|--------------------------------------|--|--------------------|-------------------|
| <b>Scan Duration</b>                 |  | From : 23 Jan 2025 | To : 28 June 2025 |
| <b>Components &amp; Abbreviation</b> |  |                    |                   |
| <b>A1</b>                            |  | Abutment 1         |                   |
| <b>A2</b>                            |  | Abutment 2         |                   |
| <b>LHS</b>                           |  | Left Hand Side     |                   |
| <b>RHS</b>                           |  | Right Hand Side    |                   |

### 3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



### 3.2 VH Displacement from 23 Jan 2025 To 28 June 2025

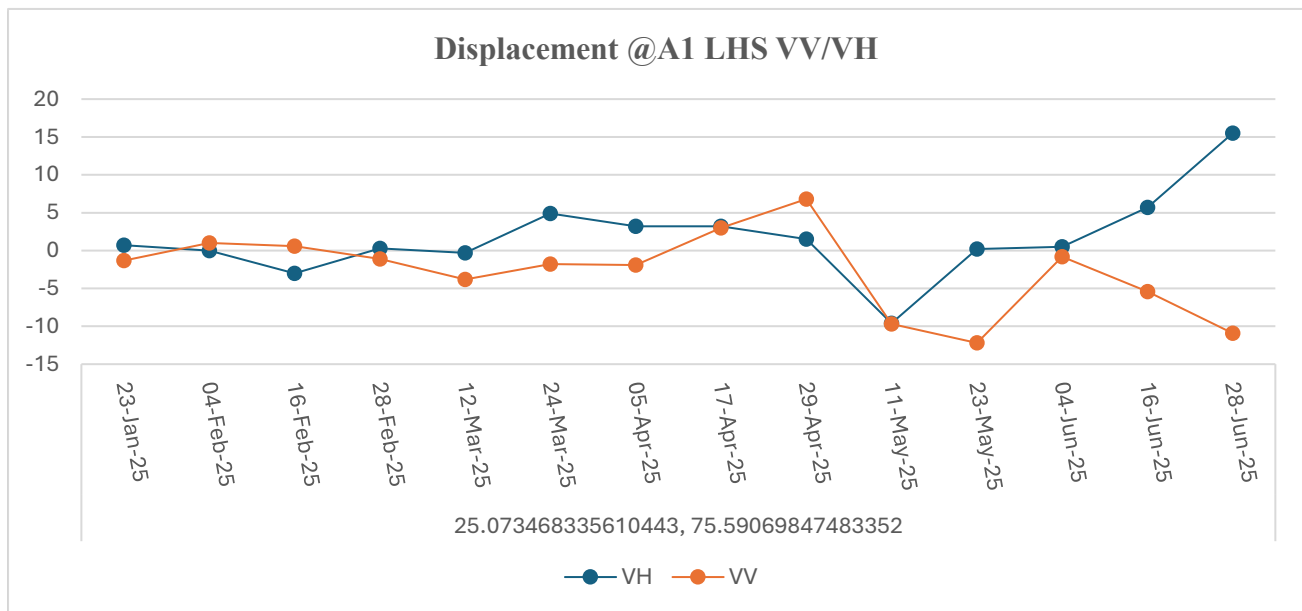




### 3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.073468335610443, 75.59069847483352)



#### Key Observations:

##### Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement exhibits clear variability with a transition from mild uplift to pronounced settlement over the monitoring period (Jan–Jun 2025). VV values range approximately from +6 mm to –12 mm.
- **Horizontal Displacement (VH)** Horizontal displacement shows moderate to high variability with an overall increasing trend. VH ranges roughly between –3 mm and +15 mm, transitioning from small oscillations around zero in the early months to strong positive lateral displacement by late June, suggesting progressive lateral activation at the A1 LHS location.

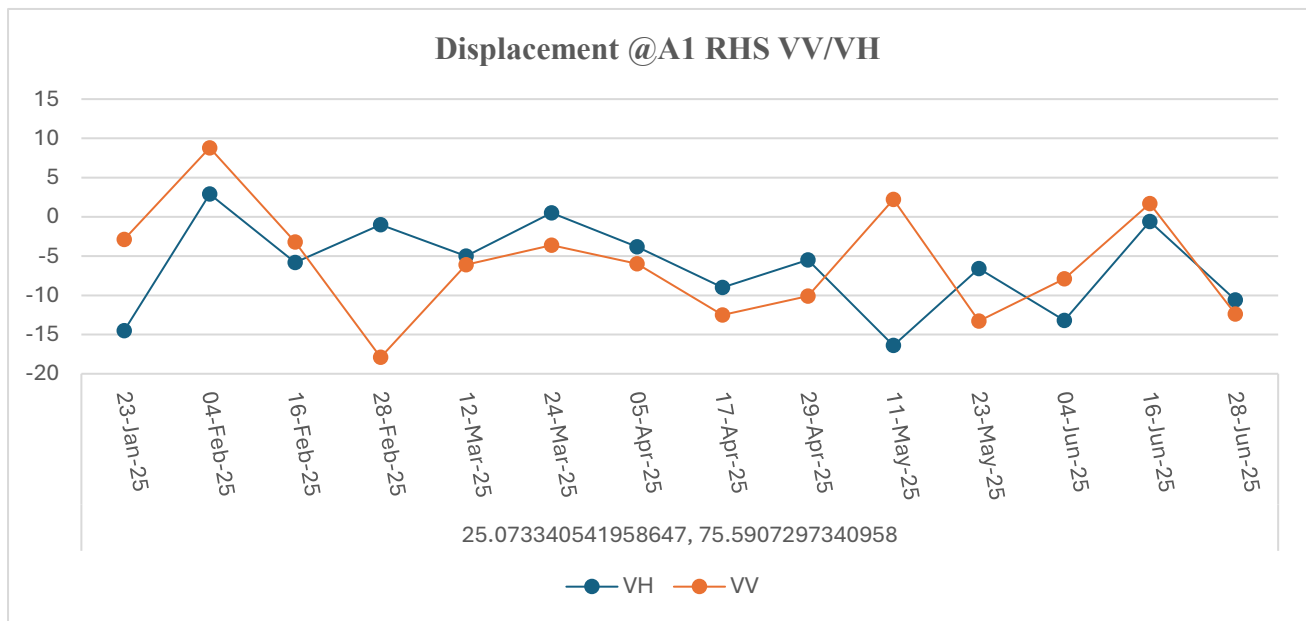
##### Critical Events & Anomalies:

- **February 2025:** Both VH and VV remain close to neutral values ( $\pm 2$  mm), indicating a relatively stable deformation phase with no dominant vertical or horizontal response.
- **June 2025:** VH increases rapidly from +1 mm to +15 mm, while VV remains strongly negative (–5 mm to –11 mm), indicating dominant lateral displacement developing alongside continued vertical settlement.

### 3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.073340541958647, 75.5907297340958)



#### Key Observations:

##### Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement shows moderate variability with predominantly negative values, ranging roughly between +2 mm and –16 mm.
- **Vertical displacement (VV)** Vertical displacement exhibits high variability with alternating uplift and settlement phases throughout the monitoring period (Jan–Jun 2025). VV values range approximately from +8 mm to –18 mm, indicating an unstable vertical response with no sustained long-term trend toward either uplift or settlement.

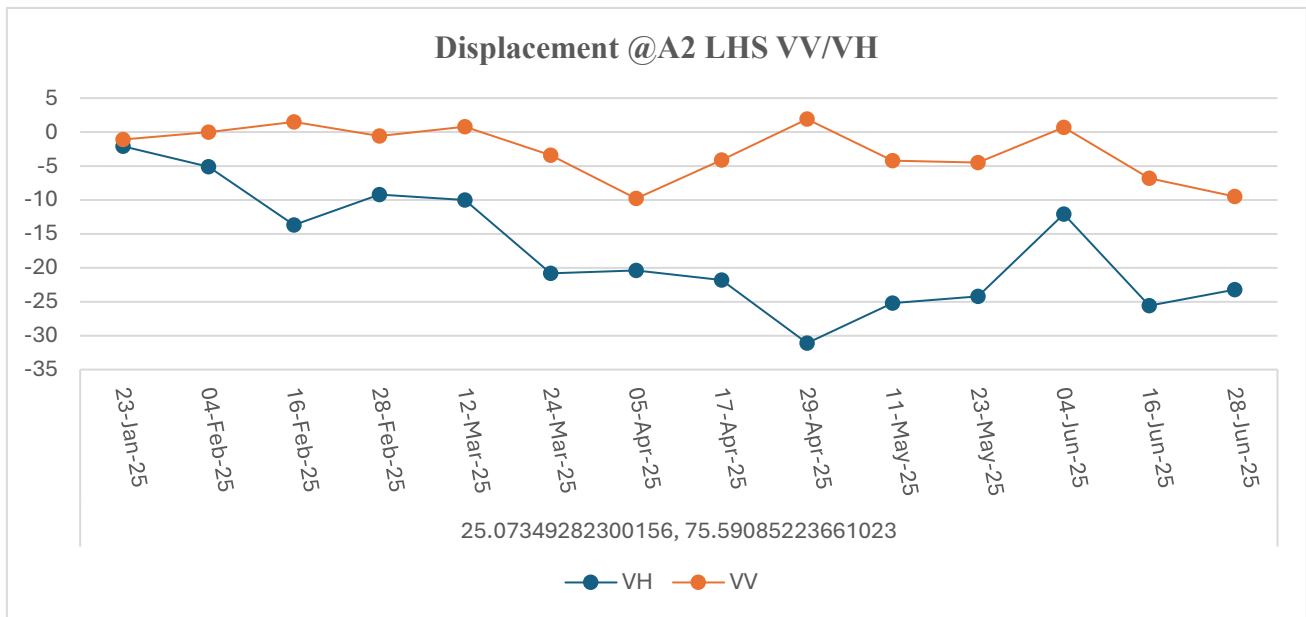
##### Critical Events & Anomalies:

- **February 2025:** VV reaches a local positive peak (+8 mm), while VH also increases to +2 mm, indicating a short-lived coupled uplift event affecting both vertical and horizontal components.
- **June 2025:** Both components decline again (VH –10 mm, VV –12 mm), confirming continued deformation activity with no sustained recovery.

### 3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.07349282300156, 75.59085223661023)



#### Key Observations:

##### Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement exhibits strong negative dominance with high variability, ranging roughly between  $-5$  mm and  $-32$  mm.
- **Vertical Displacement (VV)** Vertical displacement remains predominantly negative throughout the monitoring period (Jan–Jun 2025), indicating overall vertical settlement at the A2 LHS location.

##### Critical Events & Anomalies:

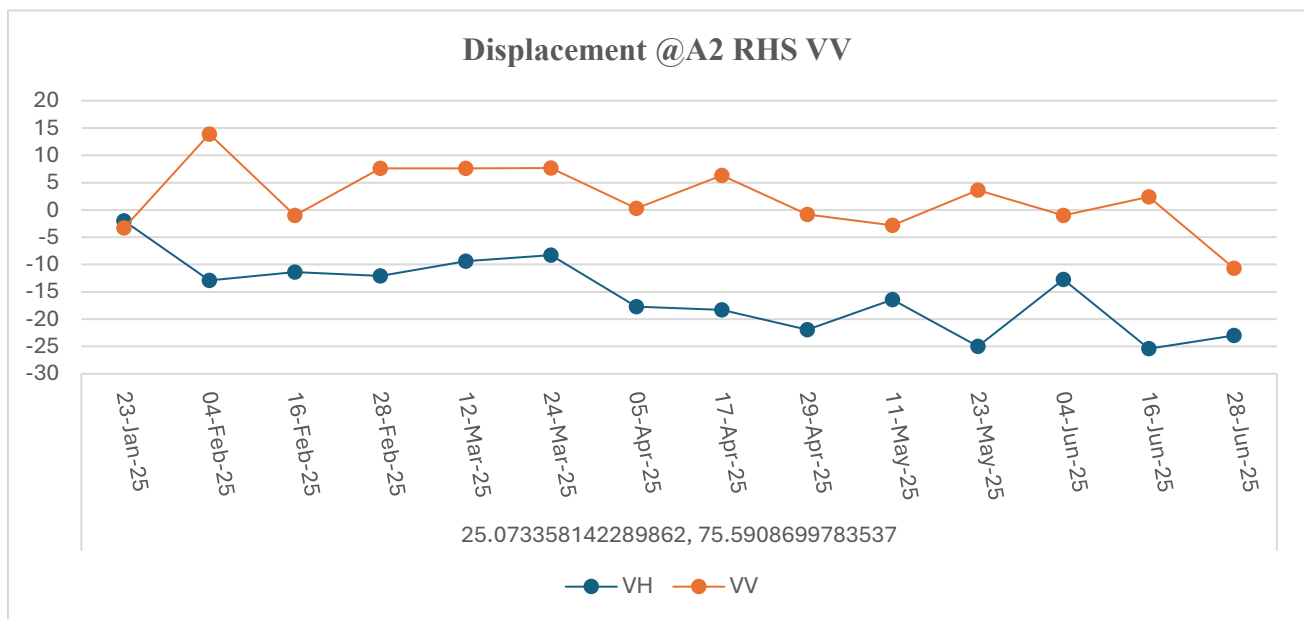
- **February 2025:** VH stabilizes briefly around  $-10$  to  $-11$  mm, while VV shows small uplift ( $+1$  to  $+2$  mm), suggesting temporary lateral stabilization under minor vertical uplift.
- **May 2025:** VH improves marginally to  $-25$  mm, while VV remains negative ( $-5$  mm), suggesting partial lateral recovery without vertical stabilization.
- **June 2025:** VH declines again to  $-24$  mm, while VV falls to  $-9$  to  $-11$  mm, confirming continued settlement toward the end of the monitoring period.



### 3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.073358142289862, 75.5908699783537)



#### Key Observations:

##### Displacement Magnitude Patterns:

- **Vertical (VV) displacement** Vertical displacement shows moderate variability with an overall downward tendency across the monitoring period (Jan–Jun 2025). VV values range approximately from +14 mm to –12 mm.
- **Horizontal (VH) displacement** Horizontal displacement remains consistently negative with high variability, ranging roughly between –10 mm and –27 mm. VH exhibits repeated deep negative excursions, indicating persistent lateral settlement in a single dominant direction throughout the observation period.

##### Critical Events & Anomalies:

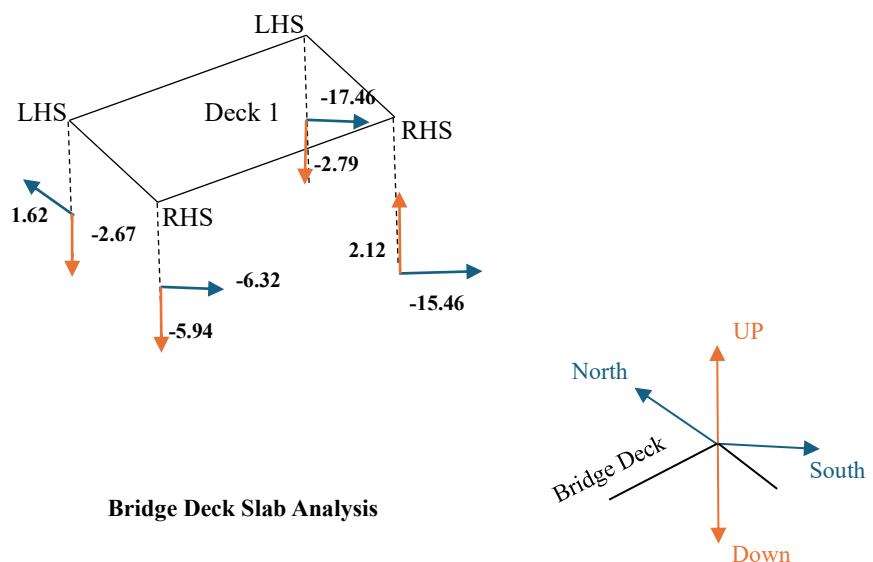
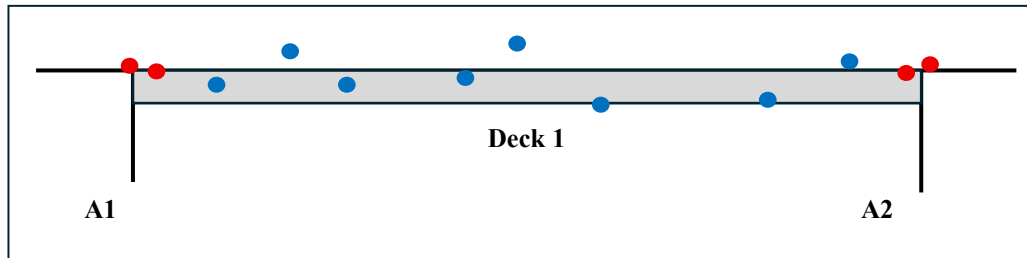
- **March 2025:** VV remains moderately positive (+5 mm), while VH declines further to –17 mm, marking a decoupled deformation phase dominated by lateral settlement.
- **June 2025:** VH improves to –12 mm, while VV decreases again to –2 mm, suggesting partial lateral recovery without sustained vertical stabilization. VV drops to –10 to –12 mm, while VH again declines to –24 mm, confirming continued settlement toward the end of the monitoring period.



## ANALYSIS

### 1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



#### Vertical Displacement Behaviour:

- The bridge deck slab shows moderate but clearly non-uniform vertical displacement, indicating differential vertical response between the LHS and RHS supports. The vertical displacement pattern confirms uneven load distribution and differential support behaviour across the deck width.

#### Horizontal Displacement Behaviour:

- Horizontal displacement displays strong directional variation across the deck, clearly reflecting torsional deformation of the bridge deck slab. The maximum horizontal displacement is recorded at the RHS lower region, reaching approximately  $-15.46$  mm, representing dominant lateral translation on the RHS. The reversal in displacement direction between LHS and RHS strongly suggests torsional rotation about a longitudinal axis of the deck.

#### CONCLUSION:

- The bridge deck slab exhibits a torsion-governed deformation pattern, characterized by moderate vertical settlement combined with significant asymmetric horizontal displacement. Overall, the observed behaviour warrants continued structural monitoring, particularly to track torsional effects under varying load and environmental conditions